

Verse Mastery

A hands-free voice agent that scores what you said,
not what it hoped you would say.

github.com/kvadney-insomniac/verse-mastery

AssemblyAI

Cloudflare Workers

MIT

THE MOMENT THIS IS BUILT AROUND

One recitation. Two scores.

"The Lord is my..." The Lord is my shepherd, I shall not want.

NAIVE COMPARISON

14%

Walks transcript and text in step.
Desynchronizes on word one and never
recovers.

SEQUENCE ALIGNMENT

100%

Omissions expensive, insertions nearly
free. Still fails a genuinely incomplete
recital at 50%.

A closed voice loop, with your hands on the wheel

- ▶ It **speaks a reference**. You recite the passage from memory.
- ▶ It **scores the recitation**, tells you how you did, and moves on.
- ▶ **No screen interaction at all**. Built for the car and the running trail, the two places a memorization app has your attention and cannot have your hands.
- ▶ Originally written for **Acts 2 Network Berkeley**; much of the app code is theirs, used under MIT. This submission is about the scorer, the ASR architecture, and the AssemblyAI integration.

Three failures nothing tunes around

- ▶ Chrome ends a **continuous session silently near 60 seconds**, with no error a caller can tell from a real ending. A long passage sits exactly on that line.
- ▶ On Chrome for Android **continuous is a documented no-op**. The car is a phone, so the mode's whole use case is the one where the flag does nothing.
- ▶ Restarting a session that ended gets **rate limited permanently**.

Record once, transcribe once. All three disappear, and the wait lands inside a gap the design already had.

Using AssemblyAI correctly, not partially

Vocabulary boosting: yes

A server-side word list of proper nouns and archaic forms the engine lacks.
Melchizedek, Zerubbabel, thy, steadfast.
Passed as **word_boost** with **boost_param: Low**. Fixes "cannot spell Habakkuk."

Sequence biasing: never

Biasing toward the expected text makes the engine return it whether or not it was said. In a scoring app that is a validity bug: someone who skipped verse 3 would be told they said it.

The route accepts audio and has **no field a verse could be put into**.
A rule enforced by a missing parameter cannot be forgotten by a caller in a hurry.

Scoring speech is alignment, not comparison

- ▶ A false start is the **single most common thing** a person does when reciting from memory.
- ▶ Step-wise comparison cannot survive one. **Asymmetric gap costs** can: omissions expensive, insertions nearly free.
- ▶ Disfluencies become free **by construction**, not by maintaining a filler-word list that is wrong in a new language the day you ship.
- ▶ Shipped standalone as **recital-score**, MIT, because the problem is not specific to scripture.

"Did they say the thing" is a question no ASR API answers

Every product below writes the same brittle string comparison we started with, and gets the same 14%.

- ▶ **Language learning.** Speaking exercises that must not credit a restart as a failure.
- ▶ **K-12 reading fluency assessment.** Scored aloud, at scale, by people who are not linguists.
- ▶ **Pronunciation and speech therapy.** Progress measured against a known target.
- ▶ **Voice-agent evals.** Regression testing whether an agent said what it was supposed to.

One Worker, two routes, no bundler

- ▶ Static files, React and htm off a CDN. The **only server is one Cloudflare Worker**, because a browser cannot hold an API key.
- ▶ **/api/transcribe** is a one-line provider swap: Workers AI, Groq, or AssemblyAI, by environment variable.
- ▶ **/api/speak** is the agent's voice, Deepgram Aura 2 on the same binding. The browser's own **speechSynthesis** is a 1990s formant voice, and in a mode whose only output is a voice, that is the product.
- ▶ Both routes are POST-only, capped, and raced against a deadline. An uncontrolled proxy in front of a paid API publishes the key with your billing attached.
- ▶ Client records **Opus at 16 kbps**: a 40 second recitation is about 80 KB. Hence the pre-recorded API, not the WAV-only sync endpoint.

1,040

UNIT TESTS

~80 KB

PER RECITATION

0

ACCOUNTS REQUIRED

DEMO

Live

PLACEHOLDER. Replace with a screenshot of the deployed app mid-recital, or drop this slide and let the video carry it. This is the only slide that needs the deploy.

Repository: `github.com/kvadney-insomniac/verse-mastery`

Scorer: `npm i recital-score`

Demo: pending deploy